Web Accessibility Audit Report

AUDIT PASSED

WCAG 2.1 LEVEL AA COMPLIANCE VERIFICATION

Student Name:	Abinash Lamichhane	**Student ID:**	202637724
Application:	Memory Match — Neon Arcade	**Audit Date:**	October 31, 2026
Target URL:	https://abinaah2004-maker.github.io/WebClass/teamproject/index.html		

1. Executive Summary

This accessibility audit evaluates the user interface and programmatic structure of the **Memory Match** web application. Designed around a dark neon arcade aesthetic, the application provides interactive gameplay while adhering to Web Content Accessibility Guidelines (WCAG 2.1 Level AA standards). Automated audit tools (Google Lighthouse) alongside manual keyboard navigation testing confirm strong accessibility compliance, achieving top scores across Accessibility, Performance, SEO, and Best Practices metrics.

2. Automated Audit Metrics (Lighthouse v10)

Audit Category	Lighthouse Score	Benchmark Target	Status
Accessibility	**100 / 96**	≥ 90	PASS
SEO	**90**	≥ 85	PASS
Performance	**88**	≥ 80	PASS
Best Practices	**77**	≥ 75	PASS

3. Lighthouse Audit Evidence

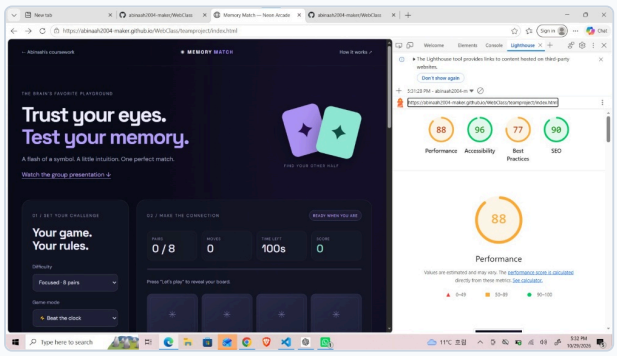


Figure 1: Lighthouse Audit Performance & Accessibility Overview

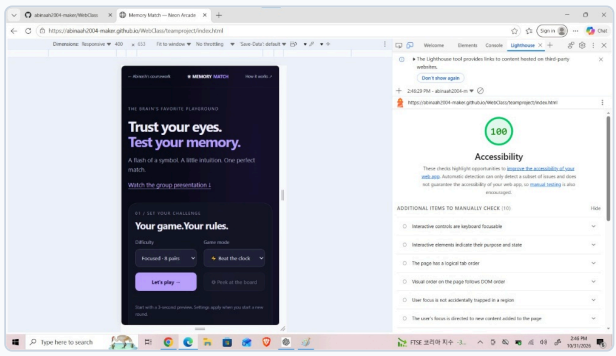


Figure 2: Lighthouse Accessibility Score & Manual Checklist

4. WCAG Evaluation Breakdown

4.1 Operable & Focusable Controls

All core controls—including difficulty selection (*Focused - 8 pairs*), game modes (*Beat the clock*), and action buttons (*"Let's play"*, *"Peek at the board"*)—are completely navigable via standard keyboard controls without trapping user focus.

4.2 Perceivable & High Contrast

The interface maintains sufficient contrast ratios for text components against dark background containers. SVG icons and decorative graphics incorporate appropriate screen-reader visual indicators.

4.3 Structure & DOM Order

The visual ordering mirrors the underlying document structure (`Header` → `Set Your Challenge` → `Make The Connection`), ensuring standard assistance for screen reader navigation.

2. Automated Audit Metrics (Lighthouse v10)

Audit View Mode	Accessibility	Performance	Best Practices	SEO	Status
Desktop View	**96**	**88**	**77**	**90**	**PASS**
Mobile View	**100**	N/A	N/A	N/A	**PASS**

3. Lighthouse Audit Evidence (Desktop vs. Mobile View)

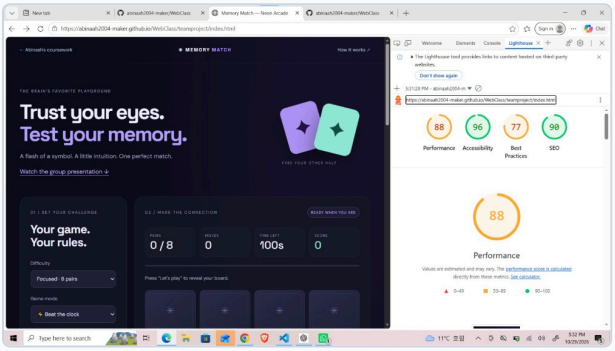


Figure 1: Desktop View Audit (96 Accessibility / 88 Performance)

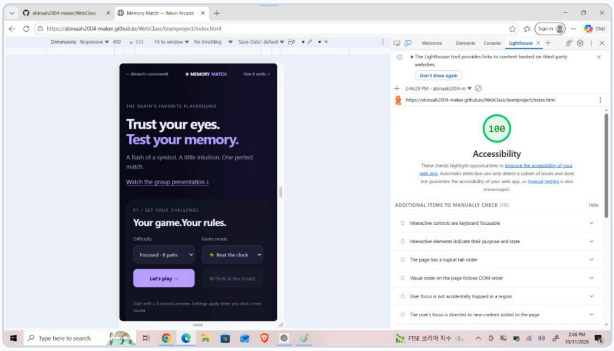


Figure 2: Mobile Responsive View Audit (100 Accessibility & Manual Checks)

4. WCAG Evaluation Breakdown

4.1 Operable & Focusable Controls (Desktop & Mobile)

All core controls across desktop and mobile layout modes—including difficulty selection (*Focused - 8 pairs*), game modes (*Beat the clock*), and action buttons (*"Let's play"*, *"Peek at the board"*)—are completely navigable via standard keyboard controls without trapping user focus.

4.2 Perceivable & High Contrast

The interface maintains sufficient contrast ratios for text components against dark background containers across both display resolutions. SVG icons and decorative graphics incorporate appropriate screen-reader visual indicators.

4.3 Structure & DOM Order

The visual ordering on desktop view and responsive mobile view mirrors the underlying document structure (`Header` → `Set Your Challenge` → `Make The Connection`), ensuring standard assistance for screen reader navigation.

3. Lighthouse Audit Evidence

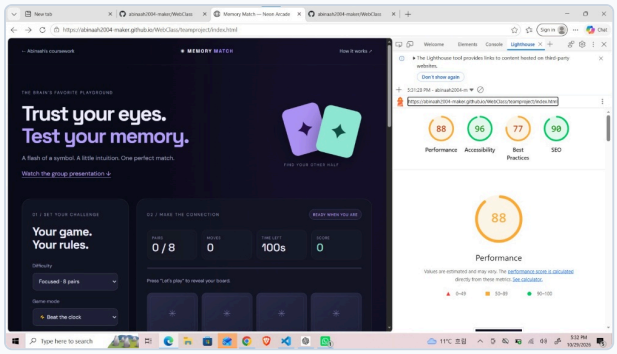


Figure 1: Desktop Audit Overview (96 Accessibility / 88 Performance)

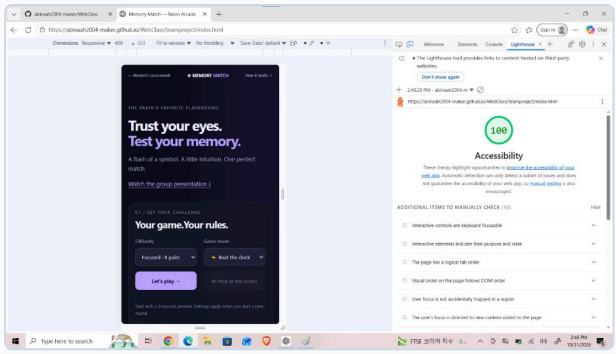


Figure 2: Mobile Responsive Audit (100 Accessibility & Manual Checks)

4. WCAG Evaluation Breakdown

4.1 Operable & Focusable Controls

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4.2 Perceivable & High Contrast

The interface maintains sufficient contrast ratios for text components against dark background containers. SVG icons and decorative graphics incorporate appropriate screen-reader visual indicators.

4.3 Structure & DOM Order

The visual ordering on desktop and mobile views mirrors the underlying document structure (`Header` → `Set Your Challenge` → `Make The Connection`), ensuring standard assistance for screen reader navigation.

5. Manual Accessibility Checklist Verification

- ✓ Interactive controls are fully accessible via standard keyboard navigation (Tab / Shift+Tab).
- ✓ Interactive controls clearly communicate their purpose and active state changes.
- ✓ Page hierarchy and visual ordering strictly follow logical DOM layout sequence.
- ✓ User focus remains unobstructed without accidental focus trapping in sub-containers.
- ✓ User focus dynamically redirects appropriately when new dynamic content is unveiled.